TMS-EEG Perturbational Complexity Index (PCI) Report

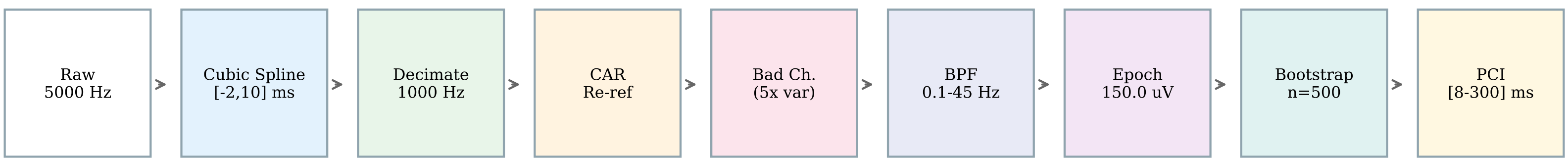
File: hert_teps_f3.vhdr

Generated: 2026-03-01 12:29

Recording Overview

Parameter	Value
Channels	62 (2 non-EEG excl.)
Sampling Rate	5000 Hz (processed at 1000 Hz)
Duration	1443.7 s (24.1 min)
Total TMS Pulses	150
Sessions Detected	3 (f3, fz, p4)
Response Markers as TMS	Yes

Processing Pipeline (per session)



PCI Results Summary

Session	Time (s)	PCI	Status	SNR	Active%	Entropy H	Epochs	Ch. Used	Bad Ch.
f3	468 - 615	**0.2599**	Unconscious	3.18	2.6%	0.176	47/50	53	9
fz	778 - 925	**0.5253**	Conscious	3.44	1.0%	0.080	40/50	53	9
p4	1101 - 1248	**0.3714**	Intermediate	2.57	1.2%	0.097	39/50	56	6

Figure 1: Full EEG Recording — TMS pulses (red) and session regions (shaded)

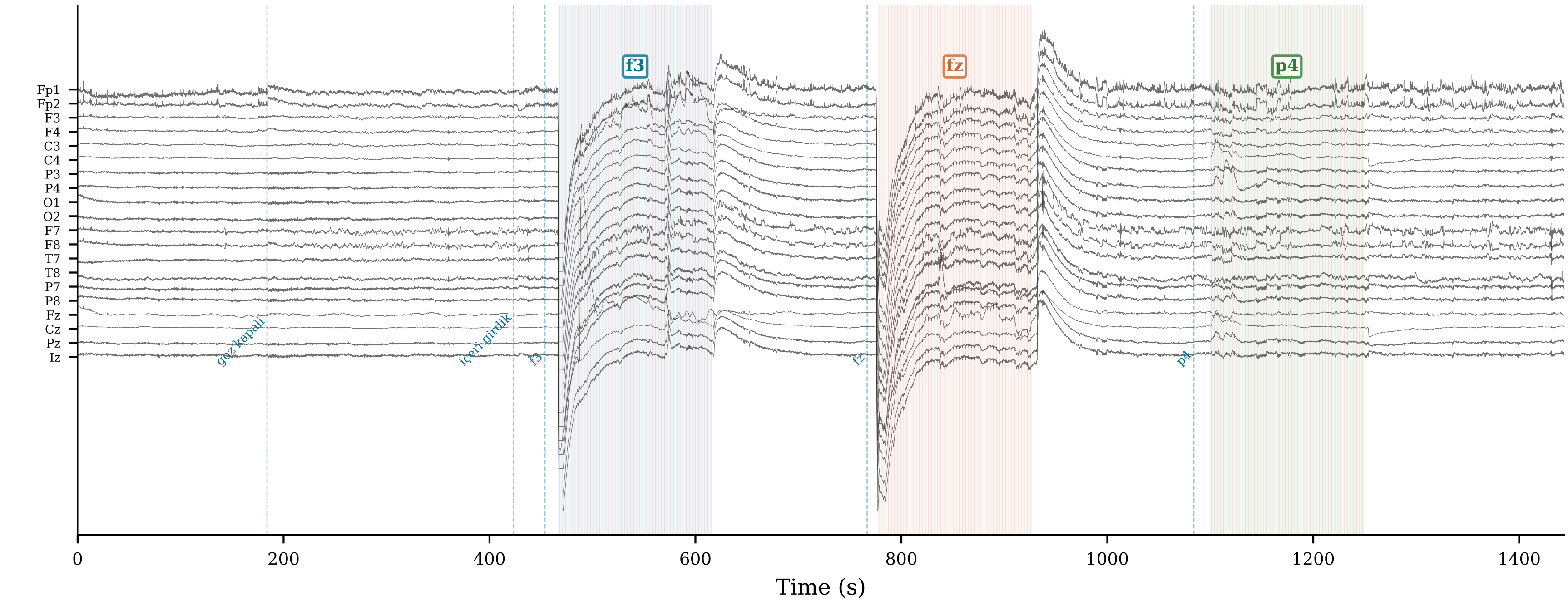


Figure 2a: PCI by Session

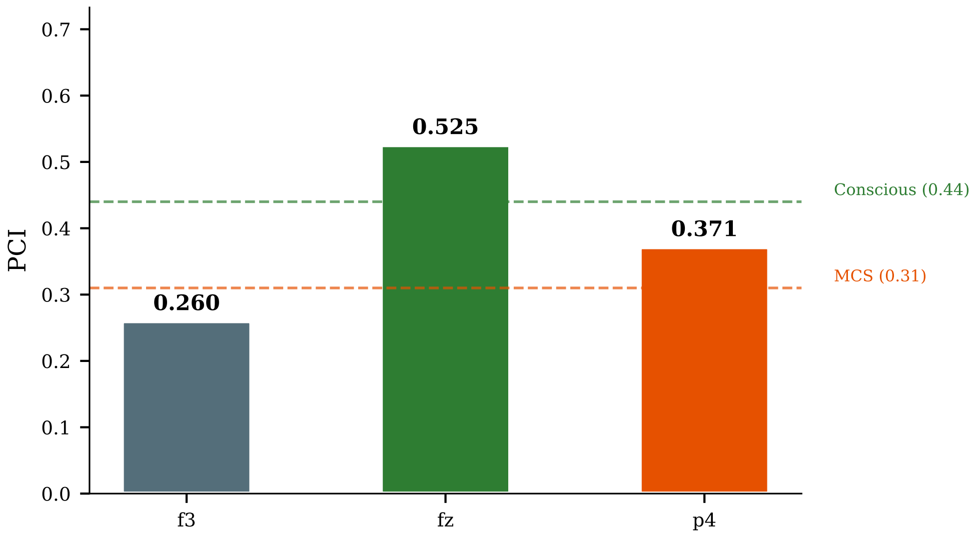
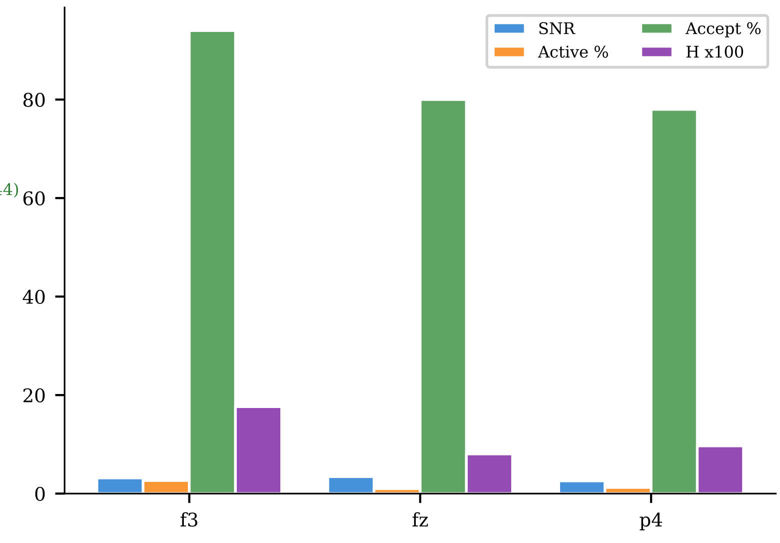
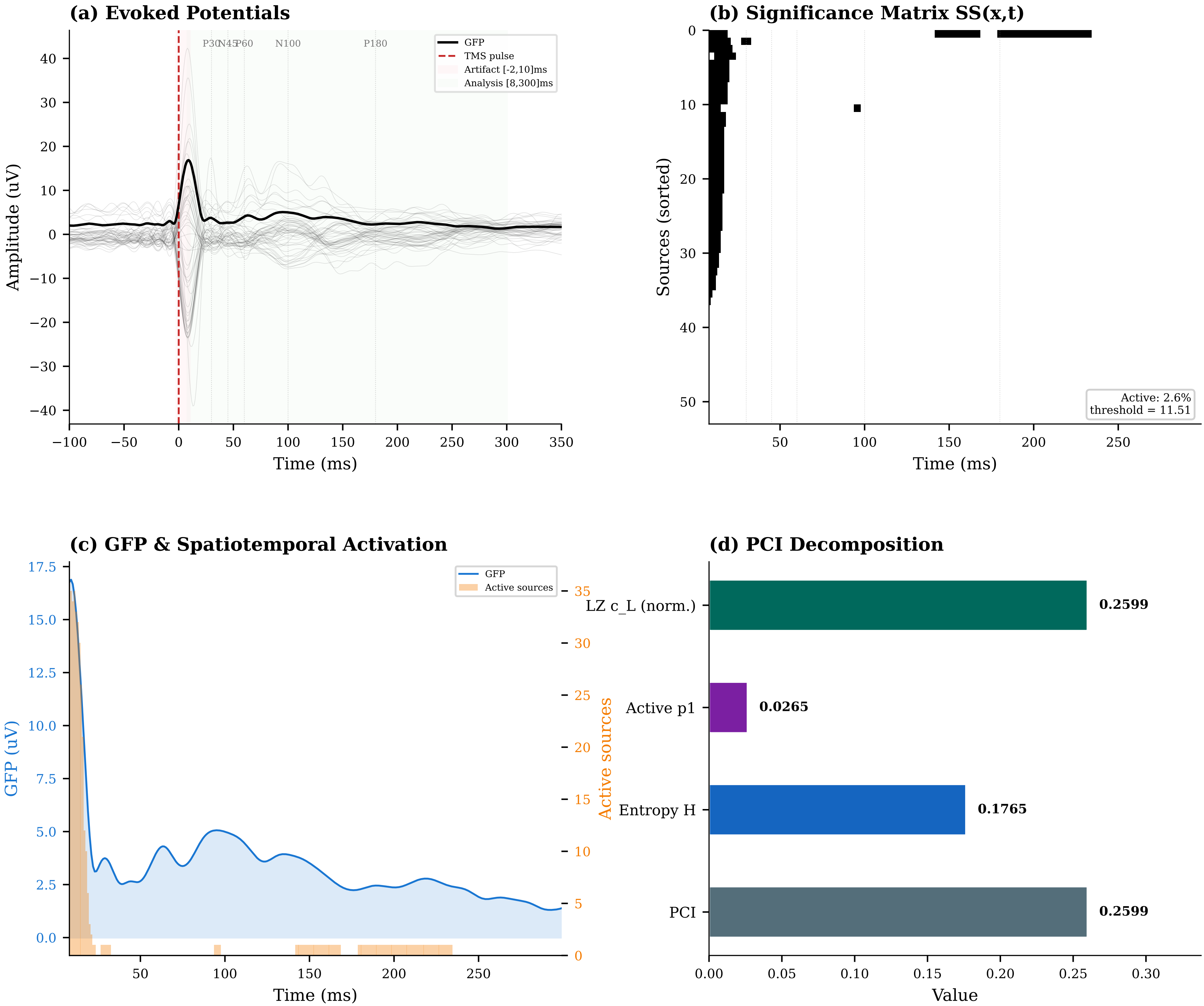


Figure 2b: Quality Metrics by Session

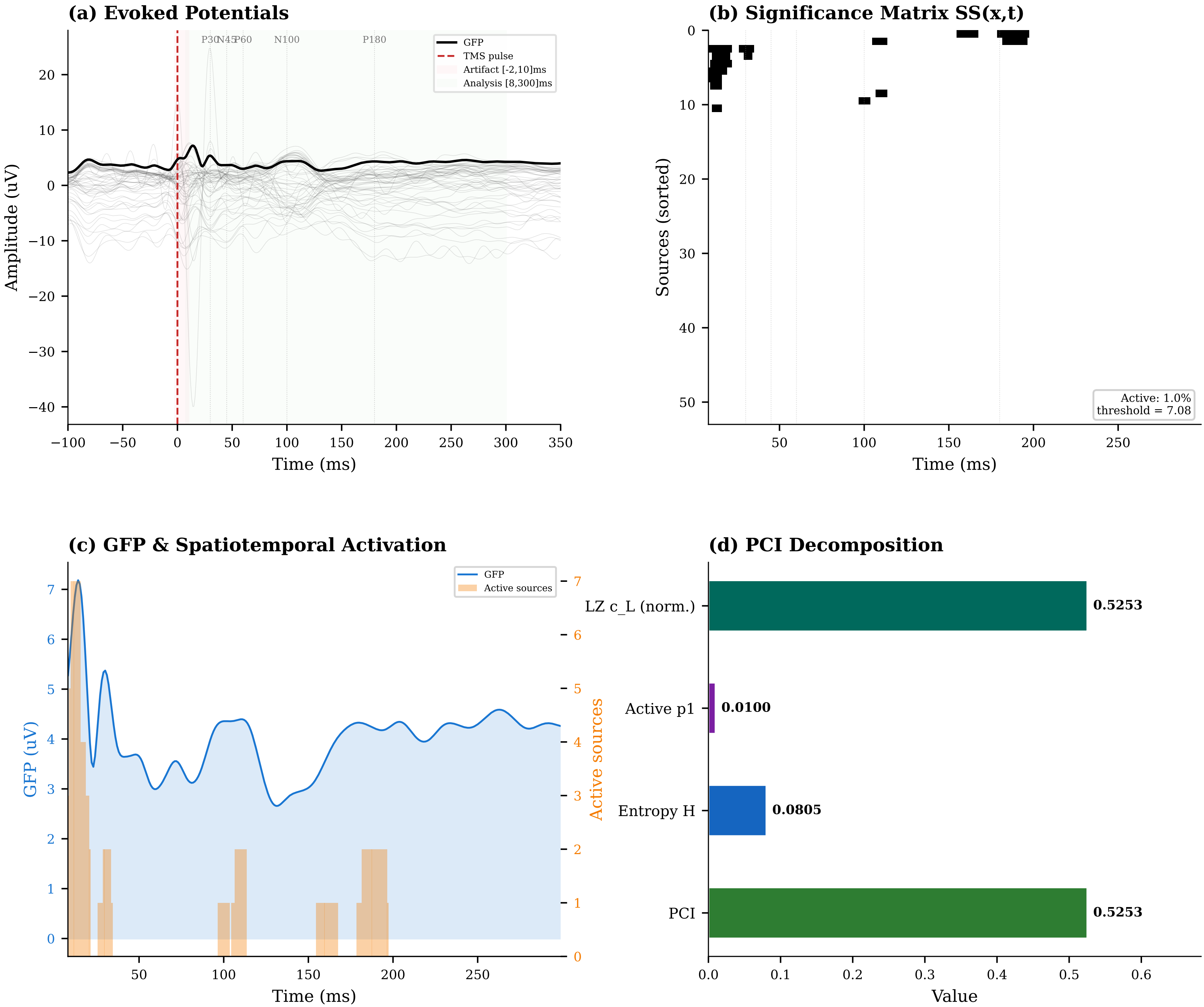


Session: f3 | PCI = 0.2599 (Unconscious) | Time: 468-615 s

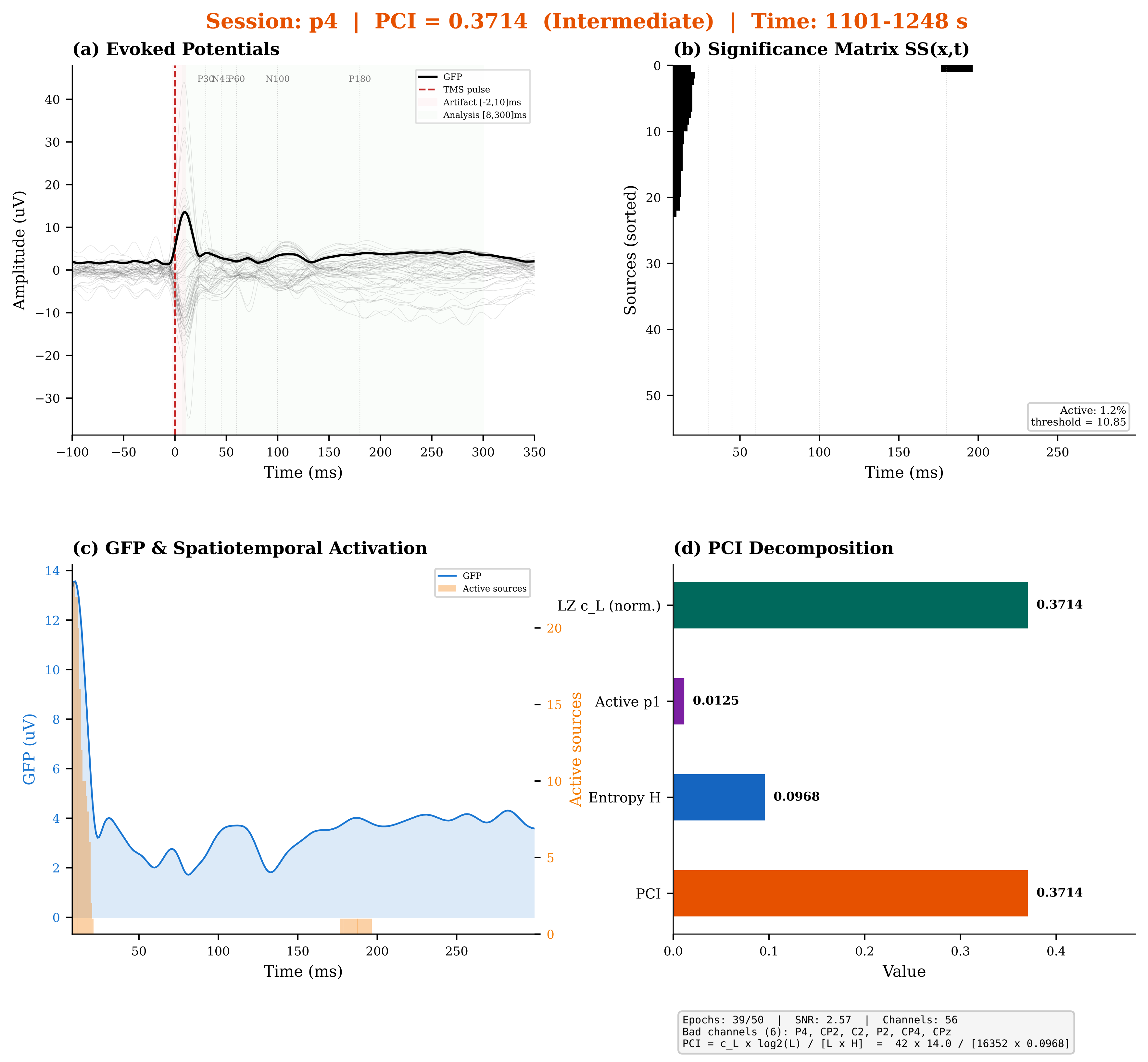



```
Epochs: 47/50  |  SNR: 3.18  |  Channels: 53
Bad channels (9): F3, Fz, Cz, FC1, FC2, F1... (+3)
PCI = c_L x log2(L) / [L x H]  =  51 x 13.9 / [15476 x 0.1765]
```

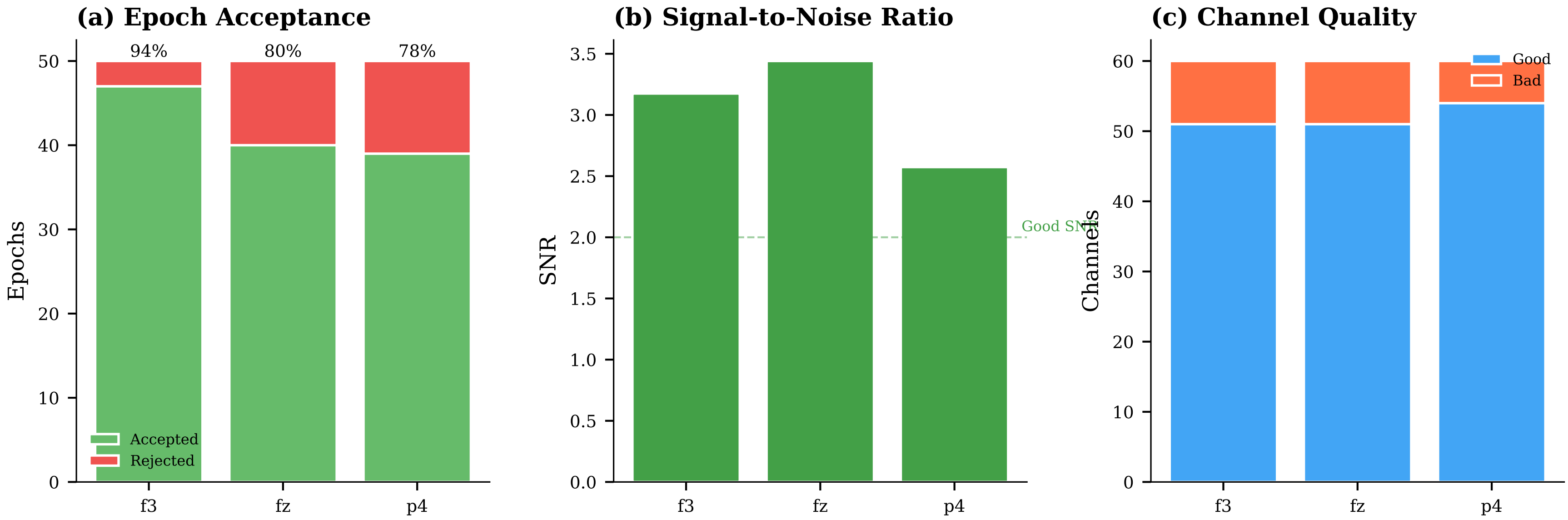
Session: fz | PCI = 0.5253 (Conscious) | Time: 778-925 s




```
Epochs: 40/50  |  SNR: 3.44  |  Channels: 53
Bad channels (9): T8, Fz, Cz, FC1, FC2, CP2... (+3)
PCI = c_L x log2(L) / [L x H]  =  47 x 13.9 / [15476 x 0.0805]
```



Session: p4 | PCI = 0.3714 (Intermediate) | Time: 1101-1248 s
(a) Evoked Potentials
GFP
TMS pulse
Artifact [-2,10]ms
Analysis [8,300]ms
P30
N45
P60
N100
P180
Amplitude (uV)
Time (ms)
(b) Significance Matrix SS(x,t)
Sources (sorted)
Active: 1.2%
threshold = 10.85
Time (ms)
(c) GFP & Spatiotemporal Activation
GFP
Active sources
GFP (uV)
Active sources
Time (ms)
(d) PCI Decomposition
LZ c_L (norm.)
0.3714
Active p1
0.0125
Entropy H
0.0968
PCI
0.3714
Value
Epochs: 39/50 | SNR: 2.57 | Channels: 56
Bad channels (6): P4, CP2, C2, P2, CP4, CPz
PCI = c_L x log2(L) / [L x H] = 42 x 14.0 / [16352 x 0.0968]



Methodology & References

Processing Pipeline: (1) Cubic spline interpolation of TMS artifact [-2, 10] ms preserving neural signal (Rogasch et al. 2017). (2) Anti-alias low-pass filter + integer decimation to 1000 Hz (2023 Brain Stimulation consensus). (3) Common Average Reference (CAR) removes shared noise. (4) Bad channel detection: channels with variance > 5x median excluded. (5) Bandpass filter 0.1-45 Hz (4th order Butterworth, forward-backward). (6) Epoching [-500, 350] ms. (7) Amplitude rejection at 150 uV peak-to-peak. (8) Non-parametric bootstrap significance testing (n=500, alpha=0.01, Global Maximum Statistics). (9) Lempel-Ziv compression of sorted binary significance matrix. (10) PCI = c_L x log2(L) / [L x H(L)].

Interpretation Thresholds (Casali et al. 2013, 208 TMS-EEG sessions in 52 subjects):
- PCI > 0.44 : Conscious range (wakefulness, dreaming, locked-in syndrome)
- PCI 0.31-0.44 : Intermediate (minimally conscious state, EMCS)
- PCI < 0.31 : Unconscious range (NREM sleep, anaesthesia, vegetative state)

Note: These thresholds were calibrated with ~5000 cortical dipoles from inverse-solution source reconstruction. Sensor-level PCI (as computed here) may require threshold recalibration (Casarotto et al. 2016).

References:
Casali AG et al. (2013) Sci. Transl. Med. 5, 198ra105.
Comolatti R et al. (2019) Brain Stimulation 12(5), 1280-1289.
Rogasch NC et al. (2017) NeuroImage 147, 16-24.